

Evaluation of Curved Raycasting-based Interactive Surfaces in Virtual Environments

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ABSTRACT

As 3D user interfaces become more popular, quick and reliable aerial selection and manipulation are desired. We evaluated a virtual curved interactive surface with controllable curvature based on raycasting. The user operates the surface by pointing using a head-mounted display and grip-type controller. We investigated the operation speed and accuracy of curved interactive surfaces under various presentation conditions. To investigate the users' operation ability for different curved conditions, we experimented with multiple surface curvature radii, including completely flat conditions. The experimental results showed that varying the curvature of the display improved the pointing accuracy by 28% and the speed by 15% over the flat surface in the most effective cases. These findings can be applied to curved interactive surfaces with mid-air pointing for 2D-style applications such as GIS and photo browsers.

Index Terms: Human-centered computing—Virtual reality; Human-centered computing—Displays and imagers

1 INTRODUCTION

Displays in a virtual environment (VE) provide a fully programmable workspace without the need for a large physical space. The advantage of such virtual workspaces is that they allow us to adopt findings from physical large screen display design [4] regarding operational efficiency. Large workspaces display a lot of information, but manipulation is difficult through the pointing interface. One solution to these problems is the design of a curved workspace [1, 2].

Although such large displays are difficult to operate, we can use the raycasting technique to interact with objects in a virtual world, which is advantageous compared with pointing in the real world [3, 5]. Therefore, we compared the operational efficiencies of flat and curved surfaces assuming that raycasting is used in VEs.

In this study, we evaluated curved virtual interactive surfaces and investigated the effects of curvature, size, and presentation distance on operational efficiency. The operation was performed using ray casting for pointing away from the user. The design parameters were analyzed based on Fitts's law, accounting for the amount of ray travel.

The paper is organized as follows. First, we present a theoretical analysis of operation time on flat and curved surfaces. Next, we describe experiments to evaluate the operational efficiency of the interactive surface with respect to pointing and clicking regarding the position and the size. Finally, we conclude this paper.

2 OPERATION TIME ANALYSIS BY FITTS'S LAW

Shupp investigated the curvature of a real large display and found that users can operate it 30% faster than flat displays [4]. In the

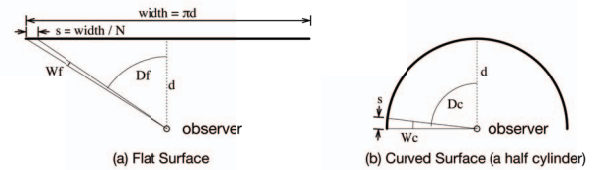


Figure 1: Models for Interactive Surfaces. The user's operation angle is controlled by the curvature radius of the virtual surface.

virtual world, we can design the display shape without any physical restrictions.

We compared the operational efficiencies of flat and curved surfaces. We conducted a theoretical analysis on the time required to point to the leftmost and rightmost selective targets using Fitts's law; $T = a + b \log_2(1 + D/W)$. For the analysis of pointing via raycasting, one controls a ray by changing the angle of the hand. Thus, we used W and D as amounts converted into angle. Let T_{end}^f and T_{end}^c be the time required for pointing to the edge targets on a flat panel and a curved panel, respectively. Figure 1(a) and (b) show the schematics of flat and curved surfaces and the operations on the assumption of one-dimensional movements. If we assume $a = 0$, the speedup by a curved surface can be expressed as $S_c^* = (T_{end}^f - T_{end}^c) / T_{end}^f = 1 - \log_2(1 + D_c/W_c) / \log_2(1 + D_f/W_f)$.

3 EXPERIMENT

To evaluate the suitability of different shapes for an immersive interactive surface, we conducted within-subject experiments to investigate the effect of display size and curvature radius on the selection of a grid of visual objects.

The participants were required to perform pointing tasks in a virtual space. A virtual interactive surface with a grid was shown to each participant at a time (Figure 2). The shape of the surface is defined from the combination of several parameters (Figure 3). The grid has the same pitch in the horizontal and vertical directions. The width was calculated from the height such that its aspect ratio tended to 16:9. We employed different sized grids to vary the difficulty of pointing.

Surfaces have the same arc length (corresponding to the width w for a flat surface) for different curvature radii r_c when the height is fixed. The viewing distance also varies (Figure 3(a)). To investigate the percentage of correct responses and operation time for the user's average operation angle, the length of the presentation distance was varied to be longer and shorter than the minimum radius of curvature.

The participant sat on a chair, put on an Oculus Rift S, and held two Oculus Touch controllers. The software for this experiment was implemented in Unity. Stimuli were provided to the participants as a colored square for each surface condition. Eight healthy adults (age: 38 ± 10 , gender: male) participated in the experiment.

The number of surface conditions is 90 ($3 \times 5 \times 2 \times 3$). A surface is chosen in order from the combination table of 4 attributes. Because the first factors vary fastest (see Figure 3(c)) in the combination table and flat conditions scatter in the surface sequence, a counterbalance on the factor of curvature radius was achieved. Stimuli

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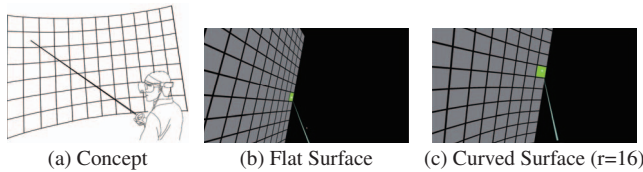


Figure 2: Interactive Surface: (a) Concept (b)(c) Participant's Perspective of Marginal Areas. The green target is the selective target.

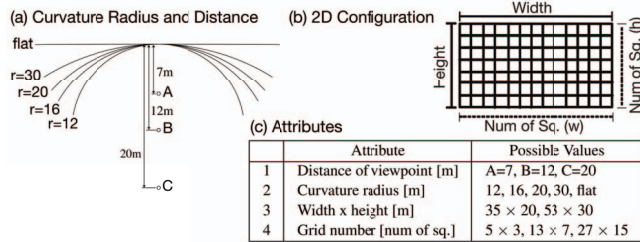


Figure 3: Surface Conditions for Experiments. (a) Curvature radii and distance are presented in all combinations. (b) The number of squares, width, and height vary as shown in the table (c).

are provided as defined in each experiment. We measured the time from the appearance of the stimulus till when the participant pressed the trigger button. If a participant pressed the trigger button while aiming the ray at a wrong square, we counted it as an error.

3.1 Experiment 1: Target Pointing without Search

The target squares repeatedly appeared in three fixed locations (left end, right end, and center as a reset position) of the center row. This experiment measured the basic sensorimotor response for raycasting on a virtual surface. The number of trials was 540 (90 surface conditions × 6 stimuli without centers) for each participant.

Figure 4 shows the plots of the percentage of correct answers vs curvature radius (r) and Table 2 shows the error ratios of experiment 1. The larger the radius of curvature, the larger the error ratio. In the case of small squares (27×15 grid), the worst percentage of correct answers is 70%, when the flat surface was operated from viewpoint A. This percentage could be improved to 98% using curved conditions ($r=12$ or 16), which was a 28% improvement. One reason for this is that squares contain large perspective distortions in the edges when the flat screen are used (Figure 2).

As for response time, the curvature radii of the shortest operation time are 20 (A), 16 (B), and 16 (C) in the case of the smallest size target. We summarized the speedup from the flat surface along with the predicted speedup obtained by Fitts's law (S_c^*) in Table 1. We confirmed that curved surfaces enable speedup in pointing operations by T-test. Fitts's law fitted well to the measured data in viewpoint C. Although the learning effect was included in the result by repeated actions, it is considered that the effect took place on all factors equally and we can regard the observed tendency as reliable.

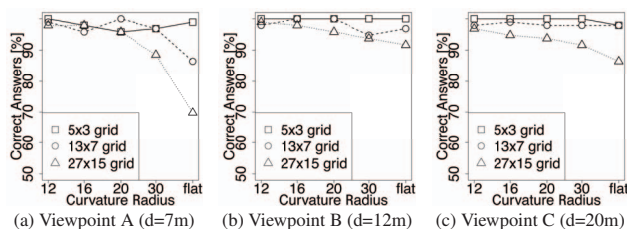


Figure 4: Percentage of Correct Answers in Experiment 1

Table 1: Speedup by Curved Surface (27×15 grid)

Viewpoint	Time by Curved Surface [sec]	Time by Flat Surface [sec]	Speedup Measured [%]	Speedup Predicted [%]
A	1.05 ± 0.24 ($r=20$)	1.20 ± 0.47	12.8*	33.6
B	0.95 ± 0.18 ($r=16$)	1.00 ± 0.30	4.9	23.7
C	0.87 ± 0.20 ($r=16$)	1.02 ± 0.46	14.6*	14.0

$a \pm b$ denotes that a is a mean and b is a standard deviation.

* denotes that it is significant at 5% level of significance by T-test.

Table 2: Average Error Ratios Obtained in Experiments 1 and 2 [%]

	$r=12$	$r=16$	$r=20$	$r=30$	flat
Experiment 1 (without Search)	1.2	1.8	2.3	4.3	8.2
Experiment 2 (with Search)	3.7	5.0	5.1	5.5	5.7

3.2 Experiment 2: Target Pointing with Search

This experiment incorporates a target seeking factor in addition to Experiment 1 in order to prevent the user from remembering the location of the target. For each surface condition, the participants had to point to three targets. A target appeared in a position from a position sequence at a time, generated randomly for each surface condition and common among all participants. The number of trials was 270 (90 surface conditions × 3 stimuli) for each participant. No reset was given in this experiment.

Table 2 shows the average error ratios of Experiment 2. The larger the radius of curvature, the larger the error ratio. This trend was consistent with that of Experiment 1.

As for response time, the observed trends were different from those observed in Experiment 1. Although we omit the details here, the flat condition was the fastest. There are two possible reasons: (a) the effect of seeking time for the out-of-sight targets is large; (b) difficulty varies between factors because each position sequence was generated randomly and short. We need to improve experiments carefully to address these issues in the next step.

4 CONCLUSION

In this research, we designed experiments to evaluate raycasting-based interactive surfaces and measured the percentage of correct answers and the operation time when surface factors were changed in a VE. The results showed that in the curved surface condition, the percentage of correct answers improves in basic and realistic situations, and the operation time shortens in the situation without target seeking activities. These findings can be applied to 2D-style applications such as GIS and photo browsers. As for future work, the experiment should be conducted to clarify the effect of out-of-sight target-seeking and normalize the difficulty of the target-seeking task.

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